

MODULE 4 APPLICATION OF DIFFERENT PROGRAMMING AND SCRIPTING LANGUAGES

The five most common programming languages in today's software developing roles are Python, Java, JavaScript, C#, and C++. Python and JavaScript are both programming languages and scripting languages. Other popular scripting languages include PHP, PowerShell, Bash, and Perl. All five of the programming languages support Windows, Linux, and macOS (which is common for modern programming languages) and have varying support of Android, iOS, embedded systems, and web browsers.

As scripting tools, Python and PowerShell support Windows, Linux, and macOS platforms. JavaScript primarily supports Web browsers and server applications while Bash supports Linux, Unix, and macOS and Perl supports Windows, Linux, and Unix.

Python : Both a programming Language and a scripting language

Python is a very versatile programming language and a scripting language. As a programming language, Python supports Artificial Intelligence, data science, test automation, process automation, developing web sites and in machine learning applications. Scripting languages support automating processes, administering systems, completing data analysis and web development. In addition to those mentioned, Python programming implements gaming applications, SQL queries, and data analytics. The large library sets, portability across platforms, and use as both a scripting tool and a programming tool, results in Python being a top programming language in use today. Python as a programming language will support Window, Linux, macOS, and limited android, iOS, embedded systems, and web browser platforms. Python programming skills are extensively in demand for embedded software test automation in the aerospace industry.

Benefits of Programming languages over scripting.

Programming languages provide better scalability in complex applications than scripting languages. Software programs efficiently utilize computing resources with less execution overhead and less memory utilization. Programming languages typically are tightly typed preventing run time errors that occur with scripting languages which are dynamically typed.

Scripting languages interpretation occurs at run time which results in slower execution times whereas, programming languages are compiled and run as machine language which results in significantly faster execution time.

Server-side Scripting languages : Java Script and PHP

Server-side scripting languages run on the webserver providing user requests and form submission authentications, database interactions, and provide responses to browsers and mobile applications.

MODULE 4 APPLICATION OF DIFFERENT PROGRAMMING AND SCRIPTING LANGUAGES

JavaScript contributes heavily to a major portion of web development applications as well as backend server side processing of requests, querying databases, and generating content. **PHP** is another server side scripting language commonly used for ecommerce websites, user authentication, and database interactions.

Server-side scripting authentication either supports or prevents a user session. Checking credentials for documented end users is an authentication process that promotes web security. Authentication failures result in user sessions failures.

Java, C#, and C++ programming languages.

Java and C# programmers develop heavily for enterprise applications. In addition to enterprise applications, Java programmers are in heavy demand in the development of Financial system applications, while cloud applications and gaming leverage C++ programmers.

In applications where real time execution is critical such as aerospace and automotive industries C++ is employed. C++ interacts with hardware making C++ an excellent choice for HW/SW abstractions layers in embedded systems.

PowerShell, Bash, and Perl scripting uses vary among the scripting tools.

DEVOPS and continuous integration/continuous delivery(CI/CD) applications utilize PowerShell for automating tasks common to aerospace software development applications. A tool utilized extensively for automating command line tasks on both Linux and Unix systems is Bash. A powerful tool for manipulating txt files and data analysis of logged data is Perl.

Programming preference summary:

My programming preference is Python for its wide range of programming applications as well as scripting applications. I prefer Python as Python programmers are in heavy demand within aerospace and automotive industries.